RAJALAKSHMI ENGINEERING COLLEGE

Muralikarthik V 250701458-2025-CSE

M2

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|---|---|
| **Started on** | Friday, 21 August 2026, 4:22 PM |
| **State** | Finished |
| **Completed on** | Friday, 21 August 2026, 4:30 PM |
| **Time taken** | 7 mins 54 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main()
{
    int m,x;
    scanf("%d",&m);
    int arr[m];
    int low=0;
    int high=m-1;
    for (int i=0;i<m;i++)
    {
        scanf("%d",&arr[i]);
    }
    scanf("%d",&x);
    while(low<=high)
    {
        int sum=arr[low]+arr[high];
        if(sum==x)
        {
            printf("%d\n%d",arr[low],arr[high]);
            return 0;
        }
        else if(sum<x)
        {
            low++;
        }
        else
        {
            high--;
        }
    }
    printf("No");
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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